

Microplastic emission flows

Anthroposphere compartments

Macroplastic emission flows

Environmental comp.

Pre- and post-consumer

Pre-consumer processes

Waste coll. and recycling

Consumption

Consumer packaging

Non-cons. packaging

Construction

Automotive

Agriculture

PCCP

Hygiene products

Clothing

Household textiles

Other products

Fishing gear

Intermediate comp.

On-the-go consumption

Litter

Dumped waste

Unintended C&D waste

Storm and waste water (MP)

Indoor air (MP)

Air (MP)

Compost and solid digestate (MP)

Incineration

Landfill

Environmental comp.

Soil (MP)

Sludge amended soil (MP)

Coastal and ocean water (MP)

Fresh water (MP)

Soil

Sludge amended soil

Coastal and ocean water

Fresh water

Litter

Dumped waste

Storm and waste water

Compost and solid digestate

The diagram illustrates the anthroposphere compartments and their interactions with environmental compartments. It is divided into three main sections: Microplastic emission flows, Anthroposphere compartments, and Macroplastic emission flows.

Microplastic emission flows

- Environmental comp.**
 - Soil (MP): 350 ± 160 t
 - Sludge amended soil (MP): 231 ± 88 t
 - Coastal and ocean water (MP): 1.46 ± 1.6 t
 - Fresh water (MP): 23.5 ± 16 t
- Intermediate comp.**
 - Storm and waste water (MP): 231 ± 88 t, 65.2 ± 44 t, 18.2 ± 14 t
 - Indoor air (MP): 92.9 ± 96 t
 - Air (MP): 1.46 ± 1.6 t, 164 ± 130 t, 5.03 ± 5.4 t
 - Compost and solid digestate (MP): 57.8 ± 22 t

Anthroposphere compartments

- Pre- and post-consumer**
 - Pre-consumer processes: 0.287 ± 0.055 t, 13.9 ± 6.9 t, 33.1 ± 14 t
 - Waste coll. and recycling: 3.7 ± 1.6 t, 2.31 ± 1.1 t, 0.00106 ± 0.0011 t
- Consumption**
 - Consumer packaging: 1360 ± 620 t, 55400 ± 33000 t, 259 ± 140 t
 - Non-cons. packaging: 2000 ± 1500 t, 71.9 ± 46 t
 - Construction: 9.42 ± 9.5 t, 537 ± 400 t
 - Automotive: 0.00298 ± 0.0015 t
 - Agriculture: 0.905 ± 0.62 t, 1.89 ± 1.2 t, 42 ± 21 t
 - PCCP: 0.0759 ± 0.049 t
 - Hygiene products: 4.24 ± 2.8 t, 757 ± 580 t
 - Clothing: 488 ± 180 t, 75 ± 54 t, 180 ± 160 t
 - Household textiles: 17.9 ± 5.7 t, 2.55 ± 1.9 t, 51.1 ± 45 t
 - Other products: 0.494 ± 0.3 t, 21.2 ± 16 t, 2.84 ± 1.4 t
 - Fishing gear: 7.47 ± 5.8 t

Macroplastic emission flows

- Intermediate comp.**
 - On-the-go consumption: 8630 ± 6100 t
 - Litter: 2440 ± 2000 t, 2.36 ± 1.9 t, 15.8 ± 14 t
 - Dumped waste: 473 ± 180 t
 - Unintended C&D waste
 - Storm and waste water: 173 ± 180 t, 23.2 ± 18 t
 - Compost and solid digestate: 46.4 ± 40 t
- Environmental comp.**
 - Soil: 3060 ± 2000 t
 - Sludge amended soil: 173 ± 180 t
 - Coastal and ocean water: 2.36 ± 1.9 t
 - Fresh water: 39 ± 28 t

The diagram shows the flow of materials between these compartments, with arrows indicating the direction of flow and numerical values representing the quantity of material in tonnes (t). The flows are categorized by color: green for organic waste, blue for water/water treatment, orange for packaging/construction, brown for waste management, and grey for other miscellaneous flows.

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Microplastic emission flows

Anthroposphere compartments

Macroplastic emission flows

Environmental comp.

Environmental comp.

Pre- and post-consumer

Consumption

Intermediate comp.

Intermediate comp.

Environmental comp.

Environmental comp.

Soil (MP)
350±160 t

Sludge amended soil (MP)
231±88 t

Coastal and ocean water (MP)
1.46±1.6 t

Fresh water (MP)
23.5±16 t

Soil
3060±2000 t

Sludge amended soil
173±180 t

Coastal and ocean water
2.36±1.9 t

Fresh water
39±28 t

Pre-consumer processes
0.287±0.055 t
13.9±6.9 t
33.1±14 t

Waste coll. and recycling
3.7±1.6 t
2.31±1.1 t
0.00106±0.0011 t

Consumer packaging
1360±620 t
55400±33000 t
259±140 t

Non-cons. packaging
2000±1500 t
71.9±46 t

Construction
9.42±9.5 t
537±400 t

Automotive
0.00298±0.0015 t

Agriculture
0.905±0.62 t
1.89±1.2 t
42±21 t

PCCP
0.0759±0.049 t

Hygiene products
4.24±2.8 t
757±580 t

Clothing
488±180 t
75±54 t
180±160 t

Household textiles
17.9±5.7 t
2.55±1.9 t
51.1±45 t

Other products
0.494±0.3 t
21.2±16 t
2.84±1.4 t

Fishing gear

On-the-go consumption
8630±6100 t

Litter
2440±2000 t
2.36±1.9 t
15.8±14 t

Dumped waste
473±180 t

Unintended C&D waste

Storm and waste water (MP)
231±88 t
65.2±44 t
18.2±14 t

Indoor air (MP)
167±66 t
20.9±20 t

Air (MP)
1.46±1.6 t
164±130 t
5.03±5.4 t

Compost and solid diges-tate (MP)
57.8±22 t

Organic waste collection
7.47±5.8 t
1390±610 t

Incineration

Landfill

Compost and solid digestate
46.4±40 t